

MODEL QUESTION PAPER 1

S6 INSTRUMENTATION ENGINEERING

SUBJECT: POWER PLANT INSTRUMENTS

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	Define Boiler efficiency	M1.04	R
2	Mention the main flow circuits of thermal power plant	M1.04	R
3	Define Corrosion	M2.04	R
4	-----is used in boiler to measure feed water flow	M2.01	R
5	List any two types of Furnace draft control	M3.03	R
6	List the fuels used in boiler.	M3.01	R
7	Write down the mechanical parameters measured in turbine system	M4.02	R
8	State the purpose of a steam turbine governing	M4.02	R
9	Name the two Non conventional energy source	M1.01	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	Compare various non conventional energy Sources.	M1.01	U
2.	List the main components of a nuclear reactor.	M1.03	R
3.	List the main components of thermal power plant.	M1.04	U
4.	Describe local boiler drum level indicator.	M2.03	U
5.	Explain two element controls	M2.03	U
6.	Describe the flow measurement of boiler feed water	M2.01	U
7.	Explain air fuel ratio control.	M3.03	U
8.	Describe dust monitor.	M3.05	U
9.	Explain the master control for power station	M4.04	U
10.	Describe the classification of turbine	M4.01	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Differentiate the non-conventional and conventional energy sources OR	M1.01	U
IV	Explain air- flue gas flow circuit	M1.04	U
V	Explain Drum level measurement methods OR	M2.03	U
VI	Describe the measurements of impurities of water	M2.04	U
VII	Illustrate the working of combustibles analyzer OR	M3.05	U
VIII	Explain the combustion control of liquid fuel	M3.03	U
IX	Explain the lubrication system for turbo alternator OR	M4.04	U
X	List the measurements in turbine safety control system	M4.03	U
XI	Explain the effect of impurities of water OR	M2.04	U
XII	Describe the superheated steam temperature control	M2.02	U
XIII	Describe the optimization of combustion control OR	M3.04	U
XIV	Explain induced draft fan control	M3.03	U

Question Wise Analysis

Course: **POWER PLANT INSTRUMENTS – QP 1.**

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.04	R	1	2
I.2	M1.04	R	1	2
I.3	M2.04	R	1	2
I.4	M2.01	R	1	2
I.5	M3.03	R	1	2
I.6	M3.01	R	1	2
I.7	M4.02	R	1	2
I.8	M4.02	R	1	2
I.9	M1.01	R	1	2
II.1	M1.01	U	3	9
II.2	M1.03	R	3	9
II.3	M1.04	U	3	9
II.4	M2.03	U	3	9
II.5	M2.03	U	3	9
II.6	M2.01	U	3	9
II.7	M3.03	U	3	9
II.8	M3.05	U	3	9
II.9	M4.04	U	3	9
II.10	M4.01	U	3	15
III	M1.01	U	7	15
IV	M1.04	U	7	15
V	M2.03	U	7	15
VI	M2.04	U	7	15
VII	M3.05	U	7	15
VIII	M3.03	U	7	15

IX	M4.04	U	7	15
X	M4.03	U	7	15
XI	M2.04	U	7	15
XII	M2.02	U	7	15
XIII	M3.04	U	7	15
XIV	M3.03	U	7	15

Course: **POWER PLANT INSTRUMENTS – QP 1.**

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Mark Distribution

Module	Hr./Module	Marks\Module (hi\ΣHi)*123(±5%)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	18	30	3	3	3	9	2	14	8	26
2	18	30	2	2	3	9	4	28	9	39
3	18	30	2	2	2	6	4	28	8	36
4	19	31	2	2	2	6	2	14	6	22
Total	75	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	12	10
Understanding	111	90
Applying	0	0
Analysing	0	0

Evaluating	0	0
Creating	0	0
Total	123	100

MODEL QUESTION PAPER 2

S6 INSTRUMENTATION ENGINEERING

SUBJECT: POWER PLANT INSTRUMENTS

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	List moderators used in nuclear power plant.	M1.03	R
2	The efficiency of a boiler is-----	M1.04	R
3	Nuclear fusion is a process of-----	M1.03	R
4	Name the sensor used to measure temperature in a boiler.	M2.02	R
5	Name the instrument which used to operate steam boiler safely.	M2.05	R
6	ppm stands for-----	M3.02	R
7	In Zirconia oxygen analyzer, the temperature at which the probe is heated is-----	M3.05	R
8	The instrument used for measuring turbine speed is -----	M4.02	R
9	The Master Control controls -----	M4.05	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	Explain the generation of electricity in thermal power plant	M1.02	U
2.	List advantages and disadvantages of nuclear power plant	M1.03	R
3.	Describe the efficiency of thermal power plant	M1.04	U

4.	Explain the remote drum level indicator	M2.03	U
5.	Draw the diagram of steam flow measurement using orifice	M2.01	U
6.	Explain Balanced draft control	M3.03	U
7.	Write the constituents of flue gases	M3.02	U
8.	Describe the measurements in air fuel circuit	M3.02	U
9.	Describe the working of turbine	M4.01	U
10.	Write short notes on measurements of electrical parameters	M4.02	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	With a neat sketch explain the working of a typical thermal power plant. OR	M1.04	U
IV	Describe the block diagram of nuclear power plant	M1.03	U
V	Explain the internal feed water treatment OR	M2.04	U
VI	Explain with diagram five element drum level control	M2.03	U
VII	Describe the combustion control with gas fuel OR	M3.03	U
VIII	With a neat sketch explain the working of Zirconia oxygen analyzer	M3.05	U
IX	With necessary diagram explain classification of turbine OR	M4.01	U

X	Explain the control schemes for turbine	M4.03	U
XI	Explain the working of Combustibles gas (CO+H ₂) analyzer. OR	M3.05	U
XII	Explain the working of Smoke detector	M3.05	U
XIII	Describe the effects of impurities in feed water OR	M2.04	U
XIV	Explain safety interlocks for Boiler operation	M2.05	U

Question Wise Analysis

Course: **POWER PLANT INSTRUMENTS – QP 2**

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.03	REMEMBERING	1	2
I.2	M1.04	REMEMBERING	1	2
I.3	M1.03	REMEMBERING	1	2
I.4	M2.02	REMEMBERING	1	2
I.5	M2.05	REMEMBERING	1	2
I.6	M3.02	REMEMBERING	1	2
I.7	M3.05	REMEMBERING	1	2
I.8	M4.02	REMEMBERING	1	2
I.9	M4.05	REMEMBERING	1	2
II.1	M1.02	UNDERSTANDING	3	9
II.2	M1.03	REMEMBERING	3	9
II.3	M1.04	UNDERSTANDING	3	9
II.4	M2.03	UNDERSTANDING	3	9
II.5	M2.01	UNDERSTANDING	3	9
II.6	M3.03	UNDERSTANDING	3	9
II.7	M3.02	UNDERSTANDING	3	9
II.8	M3.02	UNDERSTANDING	3	9
II.9	M4.01	UNDERSTANDING	3	9
II.10	M4.02	UNDERSTANDING	3	9
III	M1.04	UNDERSTANDING	7	15
IV	M1.03	UNDERSTANDING	7	15
V	M2.04	UNDERSTANDING	7	15
VI	M2.03	UNDERSTANDING	7	15
VII	M3.03	UNDERSTANDING	7	15
VIII	M3.05	UNDERSTANDING	7	15
IX	M4.01	UNDERSTANDING	7	15
X	M4.03	UNDERSTANDING	7	15
XI	M3.05	UNDERSTANDING	7	15
XII	M3.05	UNDERSTANDING	7	15
XIII	M2.04	UNDERSTANDING	7	15
XIV	M2.05	UNDERSTANDING	7	15

Course: **POWER PLANT INSTRUMENTS – QP 2**

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Mark Distribution

Module	Hr/Module	Marks\Module (hi\ΣHi)*123(±5%)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	18	30	3	3	3	9	2	14	8	26
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3	18	30	2	2	3	9	4	28	9	39
4	19	31	2	2	2	6	2	14	6	22
Total	75	123	9	9	10	30	12	84	31	123

Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	12	10
Understanding	111	90
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100

Question Wise Analysis

Course: **POWER PLANT INSTRUMENTS – QP2**

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.04	REMEMBERING	1	2
I.2	M1.04	REMEMBERING	1	2
I.3	M2.04	REMEMBERING	1	2
I.4	M2.01	REMEMBERING	1	2
I.5	M3.03	REMEMBERING	1	2
I.6	M3.01	REMEMBERING	1	2
I.7	M4.02	REMEMBERING	1	2
I.8	M4.02	REMEMBERING	1	2
I.9	M1.01	REMEMBERING	1	2
II.1	M1.01	UNDERSTANDING	3	9
II.2	M1.03	REMEMBERING	3	9
II.3	M1.04	UNDERSTANDING	3	9
II.4	M2.03	UNDERSTANDING	3	9
II.5	M2.03	UNDERSTANDING	3	9
II.6	M2.01	UNDERSTANDING	3	9
II.7	M3.03	UNDERSTANDING	3	9
II.8	M3.05	UNDERSTANDING	3	9
II.9	M4.04	UNDERSTANDING	3	9
II.10	M4.01	UNDERSTANDING	3	15
III	M1.01	UNDERSTANDING	7	15
IV	M1.04	UNDERSTANDING	7	15
V	M2.03	UNDERSTANDING	7	15
VI	M2.04	UNDERSTANDING	7	15
VII	M3.05	UNDERSTANDING	7	15
VIII	M3.03	UNDERSTANDING	7	15
IX	M4.04	UNDERSTANDING	7	15
X	M4.03	UNDERSTANDING	7	15
XI	M2.04	UNDERSTANDING	7	15
XII	M2.02	UNDERSTANDING	7	15
XIII	M3.04	UNDERSTANDING	7	15
XIV	M3.03	UNDERSTANDING	7	15

Course: **POWER PLANT INSTRUMENTS – QP 2**

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Mark Distribution

Module	Hr/Module	Marks\Module ($\sum H_i$)*123($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	18	30	3	3	3	9	2	14	8	26
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Cognitive Level Mark Distribution

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Understanding	111	90
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100